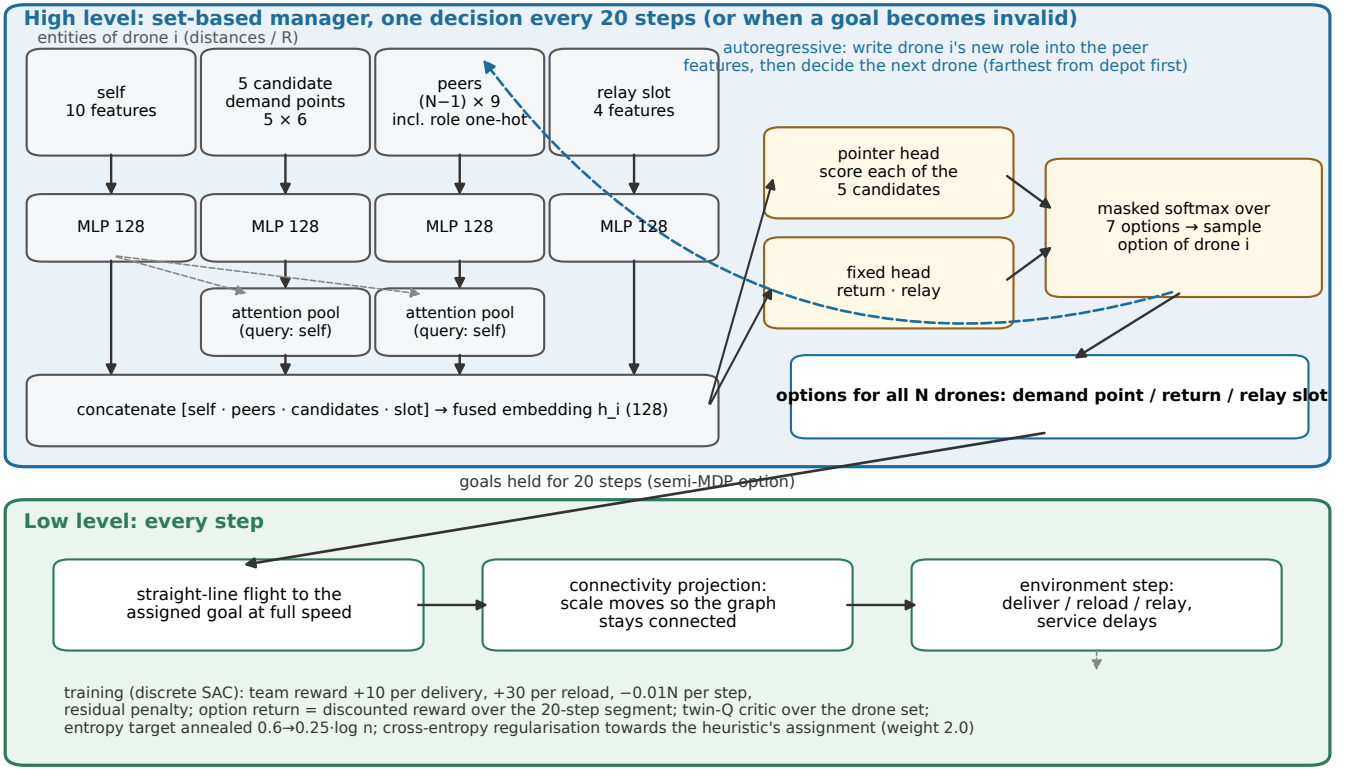
High level: set-based manager, one decision every 20 steps (or when a goal becomes invalid)
entities of drone i (distances / R)
autoregressive: write drone i's new role into the peer features, then decide the next drone (farthest from depot first)
self
10 features
5 candidate demand points
5 × 6
peers
(N−1) × 9
incl. role one-hot
relay slot
4 features
MLP 128
MLP 128
MLP 128
MLP 128
attention pool
(query: self)
attention pool
(query: self)
concatenate [self · peers · candidates · slot] → fused embedding h_i (128)
pointer head
score each of the
5 candidates
fixed head
return · relay
masked softmax over
7 options → sample
option of drone i
options for all N drones: demand point / return / relay slot
goals held for 20 steps (semi-MDP option)
Low level: every step
straight-line flight to the
assigned goal at full speed
connectivity projection:
scale moves so the graph
stays connected
environment step:
deliver / reload / relay,
service delays
training (discrete SAC): team reward +10 per delivery, +30 per reload, −0.01N per step,
residual penalty; option return = discounted reward over the 20-step segment; twin-Q critic over the drone set;
entropy target annealed 0.6→0.25·log n; cross-entropy regularisation towards the heuristic's assignment (weight 2.0)